

Simulating a quantum simulator

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We present a novel method for simulating the spin interactions of uniformly distributed particles confined to a 2D plane. By focusing on the XY model we demonstrate that our simulations are efficient and accurate in low entanglement regimes. Our method builds on previous work which isolates and simulates clusters of interacting particles to understand a larger system. We take this further with an understanding of the structure of infinite systems in order to decompose and simulate the average dynamics. By varying the regime our XY model operates within, we test the limits of our simulations and observe how edge effects interfere with our calculations. Also, we indicate areas where our method could be extended for applications in domains with higher entanglements. By considering our model within the context of an analogue quantum simulator built with ultracold polar molecules, we conclude that this work could be a useful tool for verifying the results of quantum simulators using experimentally accessible outputs. By deriving the spin $1/2$ XY model from a system of polar molecules we gain insight into the origin of our mathematical terms and how they are controlled in practice.

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Introduction

Understanding the quantum interactions of many-body systems is crucial for the development of science and technology. Superpositions of states and entanglement between particles are possible through quantum interactions and are studied within the context of many fields. Knowledge of how such systems evolve is foundational for the design and understanding of a wide number of applications. Quantum magnetism and superconductivity[8] are direct results of quantum many-body interactions, and quantum computation utilizes entangled many-body states to efficiently process information. Therefore, comprehending the dynamics of interacting quantum systems furthers our understanding of emergent physical properties and aids the effective design of quantum technologies.

Simulations are invaluable when unpicking and finding novel quantum phenomena, but modeling quantum systems is challenging. Still, there are already techniques to simulate entangled systems classically, many of which divide a larger system into smaller clusters[6] or simulate each individual particle interacting with its immediate vicinity[5]. Since the data required to store and evolve a quantum system grows exponentially with added particles, computing the dynamics in smaller parts greatly reduces the Hilbert spaces we navigate. However, these techniques are only valid when the dynamics are localized into smaller areas which only occurs at small levels of entanglement. This work builds on these methods, utilizing low levels of entanglement to simulate a many-body quantum system by decomposing it into its constituent smaller parts.

An alternative is to incorporate the natural properties of quantum systems into our models. These quantum simulators are naturally predisposed to perform the complex calculations associated with the evolution of quantum states with their own entanglement and superposition. However, the results of quantum simulators must still be validated by classical calculations. By comparing data in domains that are straightforward to simulate classically, we can be confident in the accuracy of a quantum simulator in highly entangled regimes. This work develops the tools to validate quantum simulators with classical models.

Analog quantum simulators mirror physical systems of interest with a highly tuneable experimental setup[11] and dynamics that are accessible for measurement. This work explores the many-body quantum spin interactions of particles pinned in a 2D lattice, specifically in the context of an analogue quantum simulator built with ultracold, polar $^{87}\text{Rb}^{133}\text{Cs}$ molecules[1] (hereafter referred to as RbCs). However, this work can be applied to any system of pinned particles interacting within a 2D lattice. Such simulators provide insight into quantum magnetism by mirroring electron spin interactions in a lattice with spin models. Ultracold polar molecules, like RbCs, are attractive for quantum simulation because of their intrinsic dipole moment, which can be controlled with electric fields to create long-range dipole-dipole interactions. In addition, molecules possess a rich internal structure of long-lived states from vibration and rotation and techniques for cooling and confining[9] ultracold molecules are already widespread. Additionally, the rich properties of molecules are applied in a range of fields outside of quantum simulation such as precision measurement and ultracold chemistry[10].

The correlations of 2D lattices of ultracold polar molecules have already been studied experimentally. Using quantum gas microscopy Christakis *et al*[3] realize a spin 1/2 system with ultracold polar RbCs molecules and study their correlations in a 2D lattice. This work builds on

their investigation of a 2D lattice of RbCs molecules interacting via the XY spin model.

The XY spin model has been studied comprehensively for quantum magnetism and excitation transport, nevertheless difficult regimes remain for researchers. The physics of the XY model under frustration or high levels of disorder are hard to understand, motivating the study and development of quantum simulators and other new computational techniques.

This work explores the correlations of 2D lattices interacting via the XY spin model and presents a technique to efficiently simulate such systems at low entanglement while driven by global pulses. Under these conditions, we can effectively decompose the system into smaller parts and simulate each part individually. Consequently, the Hilbert spaces we navigate are significantly smaller greatly reducing the computational load which can be used to obtain experimentally accessible values for large quantum systems. Such values are ideal for validating future quantum simulators.

The sections are structured as follows: we first rehash how the XY spin model is realized from a system of ultracold polar molecules, then we define how a two level system is driven under an external field and utilize this to probe the correlations of the system with a Ramsey pulse sequence. Next, we recap previous work, and present our computational method in depth. Finally, we display our resulting data for various lattice densities, interaction strengths and anisotropic interactions while comparing with previous computational methods.

Building a spin 1/2 system with ultracold polar molecules

In this section, we review molecular spectroscopy and the dipole-dipole interaction to construct a spin 1/2 system with pinned ultracold polar molecules. In particular, we use the rotational levels of RbCs molecules in their electro-vibrational ground state to encode a spin 1/2 system (Fig.2(b)). The physical set up described by this section is shown in Fig.2(a). As diatomic polar alkali metal dimers, RbCs molecules possess $^1\Sigma$ electronic ground states[7] so the derivations in this section can be applied to other such molecules. A diatomic molecule in its ground vibrational level of a $^1\Sigma$ electronic state has an effective Hamiltonian

$$H = H_{kin} + H_{rot} + H_{hf} + H_{ext} + H_{dd}, \quad (1)$$

where H_{kin} , H_{rot} , H_{hf} , H_{ext} and H_{dd} are contributions from the motion of the molecule, the rotation of the molecule, hyperfine structure, interactions with external fields and the dipole dipole interaction between molecules respectively.

We show that a pseudo spin 1/2 XXZ Hamiltonian can be derived from Eq.1 of the form

$$H_{XXZ} = \frac{1}{2} \sum_{i \neq j} \frac{1 - \cos^2 \theta_{ij}}{(r_i - r_j)^3} \left[\frac{J_{\perp}}{2} (S_i^+ S_j^- + S_i^- S_j^+) + J_z S_i^z S_j^z \right], \quad (2)$$

where θ_{ij} is the angle the separation vector between particles makes with the quantization axis, r_k is the position of molecule k , S^z and $S^{\pm}$ are the usual spin 1/2 operators and $J_{\perp}/J_z$ are the coupling constants. We also find expressions for the relevant terms in Eq.2.

Eq.2 is further simplified to the XY Hamiltonian, explored in this work, by setting $J_z = 0$

$$H_{XY} = \frac{1}{2} \sum_{i \neq j} \frac{1 - \cos^2 \theta_{ij}}{(r_i - r_j)^3} \left[\frac{J_{\perp}}{2} (S_i^+ S_j^- + S_i^- S_j^+) \right]. \quad (3)$$

Interestingly, both the XXZ and XY models feature an angular dependence highlighting the anisotropy of the dipole dipole interaction. This is represented graphically in Fig.1 and can be tuned by choosing the quantization axis. The overall interaction reflects how side by side dipoles repel while head to tail dipoles attract. Fig.1 also features two cones of vanishing field which make an angle of 54.7° with the quantization axis, this can be exploited to create unique interactions which reflect different (e.g. triangular) lattice geometries. The shape of the interaction is equal regardless of the choice of states which make up or spin $1/2$ system, however they do effect the magnitude of the interaction. This work focuses on the two cases displayed in Fig.1, isotropic interactions (bottom row), where there is no angular dependence in the interaction strength, and anisotropic interactions (top row). In the anisotropic case the interaction is enhanced along the z axis by a factor of two and suppressed along the diagonals displayed. The interactions are also long ranged, decaying with $1/r^3$, which is advantageous when creating highly correlated systems.

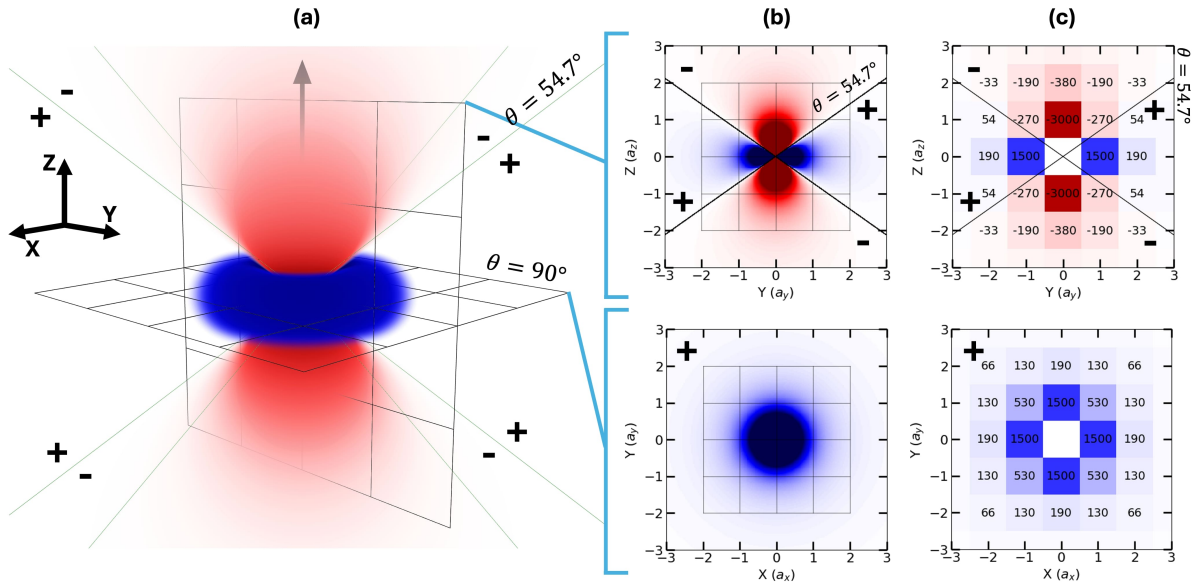


FIG. 1: Interaction strength of a single molecule with XY interactions. (a) 3D interaction strengths around a molecule with a quantization axis pointing along Z (black arrow). The density represents the interaction strength of a particle at that point and a particle at the origin. Red represents a negative interaction strength and blue a positive interaction strength, the boundary between these two, where the interaction vanishes, creates two cones with tips at the central molecule with sides 54.7° from the quantization axis. The slanted green lines trace these cones. (b) Interaction strengths around a particle in a 2D plane with different quantization axis. The cross sections each figure is derived from is labeled by the blue lines connecting to (a). The lines with vanishing interaction strength are also displayed. (c) The interaction strength of a particle projected onto a 2D lattice. Following from both cases in (b), each site's interaction strength is labeled for $J_{\perp}/2 = 1500$ Hz between nearest neighbor lattice sites.

This section inspects each term in Eq.1 separately (apart from H_{ext} and H_0 as we will consider the effect of drive fields on an arbitrary two level system in later sections and the molecules are pinned in place) before arriving at Eq.2. The dipole dipole interaction, H_{dd} , is particularly important as we derive our spin model, Eq.2, from this term.

Rotations of ultracold polar molecules, H_{rot}

Considering spinless polar molecules in their electro-vibrational ground states. Rotations of the internuclear axis mediate low-energy internal excitations of the molecules and have internal angular momentum N . The rigid spherical rotor Hamiltonian, H_{rot} , describes these excitations[7]

$$H_{rot} = B\mathbf{N}^2, \quad (4)$$

where B is the rotational constant for the electro-vibrational ground state which lies in the microwave domain. The eigenstates of Eq.(4) are labeled $|N, M\rangle$ where N is the total angular momentum and M is the projection of the angular momentum along the intranuclear axis. $|N, M\rangle$ are well known and correspond to the spherical harmonics $Y_{N,M}(\theta, \phi)$, illustrated in Fig.2(b). These eigenstates are $(2N + 1)$ fold degenerate under no external fields and obey $\mathbf{N}^2 |N, M\rangle = N(N + 1) |N, M\rangle$ and $\mathbf{N}_z |N, M\rangle = M |N, M\rangle$. In practice eigenstates with equal M mix due to the hyperfine structure of the molecule causing a level shift, Fig.2(b), however DC electric fields do not cause level crossings so we continue to use the label associated with the state obtained when the external fields are adiabatically ramped to zero.

The choice of rotational levels to encode a spin 1/2 degree of freedom allows for enhanced control of the dipole dipole interaction[1]. In this way the coupling constants may be changed or the nuclear spins could encode additional degrees of freedom[2]. In this work, we choose $|0, 0\rangle = |0\rangle$ and $|1, 0\rangle = |1\rangle$ for our derivations, the simplest level configuration which does not require microwave level dressing, to illustrate the features of the molecular Hamiltonian Eq.1 with the most clarity. We can freely choose the interaction strength in our simulations to represent other level configurations and the derivations in this section can be extended to other combinations of rotational states. [2] explores this for the interested reader.

Hyperfine structure, H_{hf}

While the rich internal structure of molecules is advantageous, it can also be difficult to control. The contribution of hyperfine structure to Eq.1 has four components: electric quadrupole, H_Q , spin-rotation, H_{IN} , scalar, H_{sc} and tensor, H_t . These are written

$$H_{hf} = H_Q + H_{IN} + H_{sc} + H_t, \quad (5)$$

These couple the nuclear spins of the atoms, $\mathbf{I}_1$ and $\mathbf{I}_2$, together and to the rotation of the molecule, $\mathbf{N}$. Note, RbCs have no unpaired electrons thus the hyperfine structure is dominated by nuclear interactions. The values of the nuclear spins are $I_{Rb} = 3/2$ and $I_{Cs} = 7/2$ [7].

Extensive knowledge of the hyperfine structure of polar molecules is not required to understand this work. Thus, we treat hyperfine structure qualitatively.

We need to prevent H_{hf} from coupling our chosen basis states $|0\rangle$ and $|1\rangle$ to other states when isolating an effective spin 1/2 system. Therefore, the hyperfine states around $|1\rangle$ must be shifted by an energy greater than the hyperfine interaction, H_{hf} . In practice we use a magnetic field to induce a Zeeman shift in the hyperfine states and to set the quantization axis.

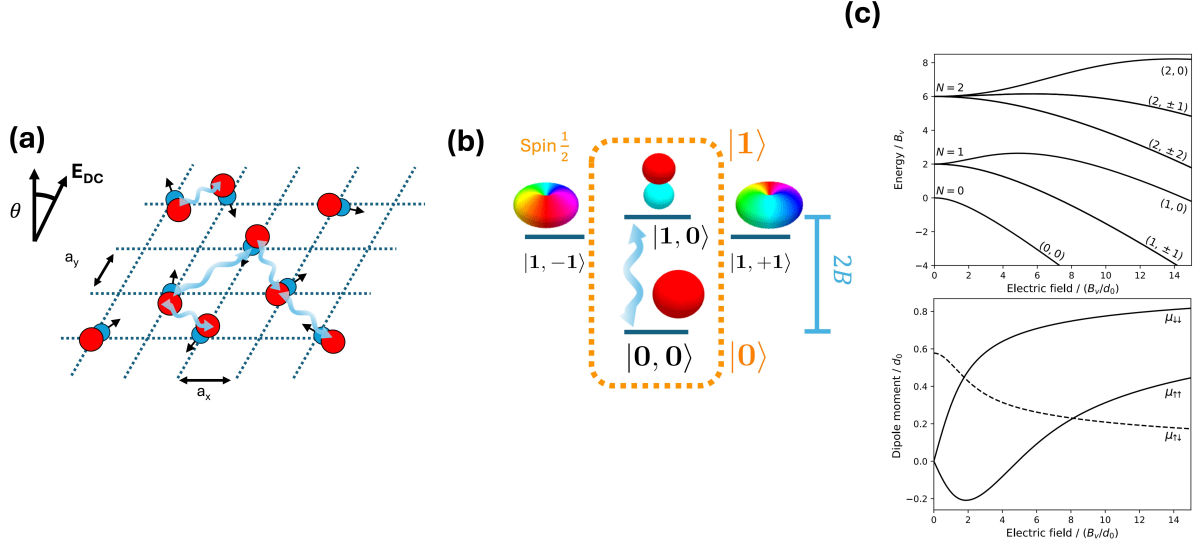


FIG. 2: Interacting ultracold alkali metal dimers. (a) Example system studied in this work. RbCs molecules pinned in a lattice with interacting (blue arrows) spins. $\mathbf{E}_{DC}$ can set the quantization axis. (b) Spin 1/2 degrees of freedom projected onto the lowest lying rotational levels of a polar molecule. Spherical harmonic plots adapted from Wikipedia. (c) Effect of a DC electric field on the rigid rotor. The Stark shifts are presented in the top plot and the matrix elements between $|0, 0\rangle$ and $|1, 0\rangle$ are illustrated in the bottom plot. Adapted from [1].

Two molecule dipole dipole interaction, H_{dd}

Now we project the dipole dipole interaction onto the spin 1/2 degree of freedom for two molecules before generalizing in later sections. We follow references [2][4] for this derivation. The dipole dipole interaction between molecules 1 and 2, H_{dd} , can be written as the contraction of two rank two tensors[2] and we can use Eq.A32 to simplify this expression

$$\begin{aligned}
 H_{dd} &= -\frac{\sqrt{6}}{4\pi\epsilon_0 R^3} T^2(\mathbf{C}) \cdot T^2(\mathbf{d}^1, \mathbf{d}^2) \\
 &= -\frac{\sqrt{6}}{4\pi\epsilon_0 R^3} \sum_{p=-2}^2 (-1)^p T_{-p}^2(\mathbf{C}) T_p^2(\mathbf{d}^1, \mathbf{d}^2)
 \end{aligned} \tag{6}$$

here $T_p^2(\mathbf{C}) = C_p^k(\theta, \phi) = \sqrt{\frac{4\pi}{2k+1}} Y_{k,p}(\theta, \phi)$, $\mathbf{d}$ describes the overall dipole operator and $\mathbf{R} = (R, \theta, \phi)$ is the separation of the molecules where θ and ϕ are the spherical angles of the diatomic molecule with respect to the quantization axis of the molecules.

In this work only the $p = 0$ components of each term is relevant in the spherical basis (since $\{-1, 0, +1\}$ correspond to the $\{\sigma^+, \pi, \sigma^-\}$ polarizations) as our chosen states have the same angular momentum projection, M . With $p = 0$ we can say

$$\begin{aligned}
 T_0^2(\mathbf{d}^1, \mathbf{d}^2) &= \frac{1}{\sqrt{6}} (\mathbf{d}_-^1 \mathbf{d}_+^2 + 2\mathbf{d}_0^1 \mathbf{d}_0^2 + \mathbf{d}_+^1 \mathbf{d}_-^2), \\
 T_0^2(\mathbf{C}) &= \frac{1}{2} (3 \cos^2 \theta' - 1).
 \end{aligned} \tag{7}$$

Here the anisotropy of the interaction is clearly presented. Substituting these into Eq.6 we obtain the dipole dipole interaction for our basis states

$$H_{dd;p=0} = \frac{1 - 3 \cos \theta'}{4\pi\epsilon_0 R^3} \left[\mathbf{d}_0^1 \mathbf{d}_0^2 + \frac{\mathbf{d}_-^1 \mathbf{d}_+^2 + \mathbf{d}_+^1 \mathbf{d}_-^2}{2} \right]. \quad (8)$$

Since $\mathbf{d}_\pm$ changes the projection of the molecular angular momentum, M , and $M = 0$ for both of our chosen states, $\langle \mathbf{d}_{\pm 1} \rangle$ vanish due to selection rules

$$H_{dd} = \frac{1 - 3 \cos \theta'}{4\pi\epsilon_0 R^3} \mathbf{d}_0^1 \mathbf{d}_0^2. \quad (9)$$

For the rest of this work we use the natural units $4\pi\epsilon_0 = 1$. We can see from Eq.9 the interactions are long ranged since they decay with $1/r^3$, advantageous for creating entangled systems.

Next, we project Eq.9 onto the internal rotational states of two interacting molecules by finding the matrix elements of Eq.9 on our basis of states. Evaluating the matrix elements of the dipole operator $\mathbf{d}_0$ in our basis $\{|0\rangle, |1\rangle\} = \{|0, 0\rangle, |1, 0\rangle\}$.

$$\langle N', M' | \mathbf{d}_p | N, M \rangle = d(-1)^{M'} \sqrt{(2N' + 1)(2N + 1)} \begin{pmatrix} N' & 1 & N \\ -M' & p & M \end{pmatrix} \begin{pmatrix} N' & 1 & N \\ 0 & 0 & 0 \end{pmatrix}, \quad (10)$$

where the 3x2 matrices are Wigner 3j symbols. Eq.10 gives the following matrix elements in lowest order of $\beta_{DC} = dE_{DC}/B$ where d is the intrinsic dipole moment and E_{DC} is the applied electric field

$$d_0 \equiv \langle 0 | \mathbf{d}_0 | 0 \rangle = \frac{\beta_{DC}}{3}, \quad d_1 \equiv \langle 1 | \mathbf{d}_0 | 1 \rangle = -\frac{\beta_{DC}}{5}, \quad (11)$$

$$d_{01} \equiv \langle 0 | \mathbf{d}_0 | 1 \rangle = \langle 1 | \mathbf{d}_0 | 0 \rangle = \frac{1}{\sqrt{3}} \left(1 - \frac{43\beta_{DC}^2}{360} \right). \quad (12)$$

Importantly, the static matrix elements in Eq.11 vanish in zero electric field while the cross terms in Eq.12 have a finite value. This is because the molecule, while featuring a dipole, has an equal chance to point in any direction and so has no net polarization. Consequently, the Ising term is eliminated in the XXZ model without an external electric field.

With Eq.11 and Eq.12 the dipole dipole interaction between two molecules can be represented as a matrix. For the basis $\{|11\rangle, |01\rangle, |10\rangle, |00\rangle\}$

$$H_{ij} = \frac{1 - 3 \cos^2 \theta_{ij}}{|\mathbf{r}_i - \mathbf{r}_j|^3} \begin{pmatrix} (d_1)^2 & 0 & 0 & 0 \\ 0 & d_0 d_1 & (d_{01})^2 & 0 \\ 0 & (d_{01})^2 & d_0 d_1 & 0 \\ 0 & 0 & 0 & (d_0)^2 \end{pmatrix}. \quad (13)$$

The off-diagonal terms facilitate an effective spin exchange between molecules, while the diagonal terms differentiate the two rotational states with respect to each other. Their corresponding values determine $J_\perp$ and J_z respectively, as seen in Eq.16 and Eq.17.

To translate Eq.13 into an effective spin model, we define a set of two particle spin 1/2 operators

$$\mathbf{S}_i^z = \frac{1}{2} (|1_i\rangle \langle 1_i| - |0_i\rangle \langle 0_i|), \quad \mathbf{S}_i^+ = |1_i\rangle \langle 0_i|, \mathbf{S}_i^- = (\mathbf{S}_i^+)^\dagger, \quad (14)$$

which satisfy the commutation relations $[\mathbf{S}_i^z, \mathbf{S}_j^\pm] = \pm \delta_{ij} \mathbf{S}_i^\pm$. Substituting Eq.14 into Eq.13 produces a two molecule effective spin Hamiltonian

$$H_{dd;ij} = \frac{1 - 3 \cos^2 \theta_{ij}}{|\mathbf{r}_i - \mathbf{r}_j|^3} \left[\frac{J_\perp}{2} (\mathbf{S}_i^+ \mathbf{S}_j^- + \text{h.c.}) + J_z \mathbf{S}_i^z \mathbf{S}_j^z + W (\mathbb{1}_i \mathbf{S}_j^z + \mathbf{S}_i^z \mathbb{1}_j) + V \mathbb{1}_i \mathbb{1}_j \right], \quad (15)$$

where

$$J_\perp \equiv 2(d_{01})^2, \quad (16)$$

$$J_z \equiv (d_1 - d_0)^2, \quad (17)$$

$$W \equiv \frac{(d_1)^2 - (d_0)^2}{2}, \quad (18)$$

$$V \equiv \frac{(d_1 + d_0)^2}{4}. \quad (19)$$

Here, the V term describes a density density interaction. The J_z term describes an Ising spin-spin interaction which promotes anti alignment of the molecules when positive. This agrees with our physical understanding of dipoles, side by side aligned dipoles repel and anti aligned dipoles attract. The $J_\perp$ term mediates the spin exchange between molecules and can mirror the super-exchange of electron spin in crystals. Molecules trade microwave excitations through this term. We explore this term in detail in this work. Finally, the W term represents a spin-density interaction. An effective magnetic field along z which acts on all other spins is generated by each molecule through this term. We generalize this equation to many molecules later in this work, however the definitions of the spin model's coefficients remain the same.

Driving two level systems

To understand the dynamics of a quantum system with reproducible experimentally accessible values, we require control over the superposition of each particle and insight into the coherences of each particles rotational levels. This may be achieved with external driving fields. In this section we detail Rabi oscillations which provide access to superposition states of choice in individual particles and Ramsey sequences which allow the correlations of particles to be measured using their coherences. Our derivations are generalized to any two level system, however to drive transitions in ultracold polar molecules we use resonant microwave radiation.

A useful way to represent the state of a two level quantum particle is the Bloch sphere, Fig.3. Each superposition of the system can be represented by a point on the Bloch spheres surface. The z axis is the population difference, which we can directly measure, and the x and y axis are the coherences between $|0\rangle$ and $|1\rangle$ which are dependent on the relative phase of the two states. Projecting the Bloch vector onto each of these axis gives their corresponding value for that state.

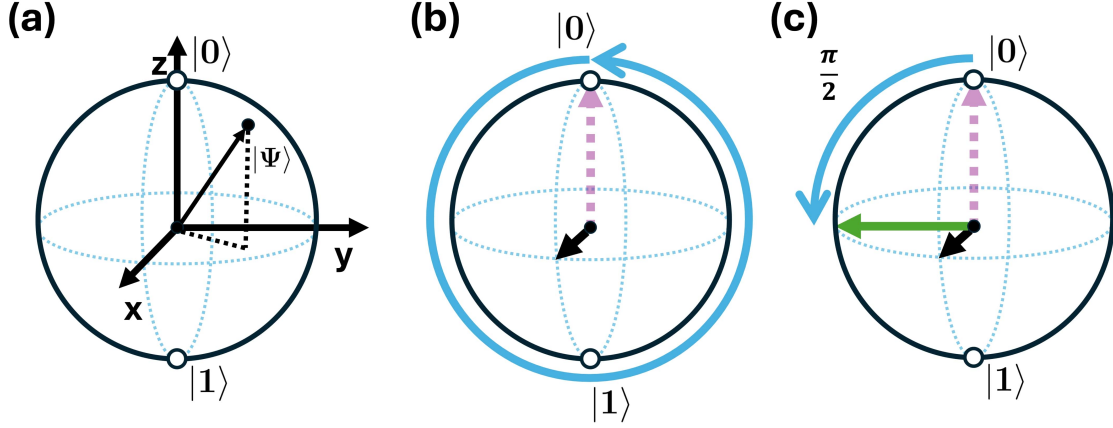


FIG. 3: The Bloch sphere and Rabi oscillations. (a) An arbitrary state $|\Psi\rangle$ represented by the Bloch vector on the surface of the Bloch sphere. The population difference between $|0\rangle$ and $|1\rangle$ is reflected on the z axis, while the x and y axis are the coherences between $|0\rangle$ and $|1\rangle$. (b) Rabi oscillations shown on the Bloch sphere. Starting from $|0\rangle$ (dotted arrow), the Bloch vector precesses about some vector (black arrow) set by the drive field's phase. (c) $\pi/2$ pulse. The dotted arrow is the initial state, the green arrow is the final state and the black arrow is the axis the rotation is driven about.

Rabi oscillations

Rabi oscillations are widely used to create superpositions and provide access to hidden variables in quantum systems. We need Rabi oscillations in this work to prepare a reproducible initial interacting quantum state. As the XY model only features spin exchange terms, molecules prepared in their ground state will not interact. Therefore, to observe quantum phenomena while utilizing global operations, we need to drive each particle into a superposition.

Consider a two level system perturbed by a small external driving field, the Schrödinger equation reflecting this interaction is[12]

$$i\hbar\partial_t |\psi\rangle = (\mathcal{H}_0 + \mathcal{H}') |\psi\rangle, \quad (20)$$

where $|\psi\rangle$ is the state vector of the two level system, $\mathcal{H}_0$ is the unperturbed Hamiltonian and $\mathcal{H}'$ is the perturbation due to an external field. Substituting the time dependent state vector $|\psi(t)\rangle = a(t)|0\rangle e^{-iE_0t/\hbar} + b(t)|1\rangle e^{-iE_1t/\hbar}$ into Eq.20 we receive a set of coupled differential equations describing $a(t)$ and $b(t)$

$$\begin{aligned} i\hbar\dot{a}(t) &= \langle 0 | \mathcal{H}' | 1 \rangle e^{-i\omega_0 t} b(t), \\ i\hbar\dot{b}(t) &= \langle 1 | \mathcal{H}' | 0 \rangle e^{i\omega_0 t} a(t), \end{aligned} \quad (21)$$

where $\omega_0 = (E_1 - E_0)/\hbar$. For an interaction with an electromagnetic driving field, $\mathcal{H}' = -\mathbf{d} \cdot \mathcal{E}$ where the electric dipole operator $\mathbf{d} = -e\mathbf{r}$ and the EM field's electric part $\mathcal{E} = \mathcal{E}_0 \cos(\phi - \omega t)$. We define the Rabi frequency $\Omega = -\langle 1 | \mathbf{d} \cdot \mathbf{E}_0 | 0 \rangle / \hbar$. Where, for real state vectors, $\Omega = -\langle 0 | \mathbf{d} \cdot \mathbf{E}_0 | 1 \rangle / \hbar$. Expanding Eq.21 using $\omega \sim \omega_0$ and neglecting $\omega + \omega_0$ terms (rotating wave approximation) we receive

$$\begin{aligned} i\dot{a}(t) &= \frac{1}{2}\Omega e^{-i\phi} e^{i\Delta t} b(t) \\ i\dot{b}(t) &= \frac{1}{2}\Omega e^{i\phi} e^{-i\Delta t} a(t), \end{aligned} \quad (22)$$

where $\Delta = \omega - \omega_0$ is the detuning of the field from the transition energy between $|0\rangle$ and $|1\rangle$. $\phi = \phi_L + \phi_q$ is the relative phase between the system and the external field and ϕ_q is the relative phase of the two level system. The explicit time dependence can be removed with a change of basis $\tilde{a} = a e^{-i(\Delta/2)t}$ and $\tilde{b} = b e^{i(\Delta/2)t}$. Now the drive field Hamiltonian is written

$$\mathcal{H}_{int} = \frac{\hbar}{2} \begin{pmatrix} \Delta & \Omega e^{-i\phi} \\ \Omega e^{i\phi} & -\Delta \end{pmatrix}, \quad |\psi\rangle = \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}. \quad (23)$$

Calculating the propagator $\exp(-i\mathcal{H}_{int}t/\hbar)$ of Eq.23 gives an answer in the form of a rotation matrix on the Bloch sphere

$$R(\Theta) = \begin{pmatrix} \cos \frac{\Theta}{2} - i \frac{\Delta}{\Omega_{eff}} \sin \frac{\Theta}{2} & -i \frac{\Omega}{\Omega_{eff}} e^{-i\phi} \sin \frac{\Theta}{2} \\ -i \frac{\Omega}{\Omega_{eff}} e^{+i\phi} \sin \frac{\Theta}{2} & \cos \frac{\Theta}{2} + i \frac{\Delta}{\Omega_{eff}} \sin \frac{\Theta}{2} \end{pmatrix}, \quad (24)$$

where $\Omega_{eff} = (\Omega^2 + \Delta^2)^{\frac{1}{2}}$ is the effective Rabi frequency and $\Theta = \Omega_{eff}t$ is the rotation angle. In the resonant driving case which is utilized in this work, $\Delta = 0$ Hz, Eq.24 simplifies to

$$R(\Theta) = \begin{pmatrix} \cos(\Omega t/2) & -ie^{-i\phi} \sin(\Omega t/2) \\ -ie^{i\phi} \sin(\Omega t/2) & \cos(\Omega t/2) \end{pmatrix}. \quad (25)$$

In the free evolution case $\Omega = 0$ Hz Eq.24 gives

$$R(\Delta t) = \begin{pmatrix} e^{-i\Delta t/2} & 0 \\ 0 & e^{i\Delta t/2} \end{pmatrix}, \quad (26)$$

which is a rotation about the z-axis. We only use resonant driving fields, so $\Delta = 0$ Hz, but the rotation about the z axis and the oscillations in the coherences during free evolution still occur in our simulations due to the dipole dipole interactions.

Fig.3(b) displays the effect of a resonant electric driving field on a single particle and is reflected mathematically in Eq.25 above. Eq.26 reflects how no component of the Bloch sphere rotates about the z-axis at $\Delta = 0$ Hz thus, for an initial state $|0\rangle$, we observe a rotation of the Bloch vector between $|0\rangle$ and $|1\rangle$ with the specific plane of rotation determined by the phase of the field ϕ .

Driven Rabi oscillations are extremely useful for state preparation and Ramsey interferometry as, by varying the time an external field perturbs a two level system, any point on the Bloch sphere becomes accessible. Consequently, if a system initially in the state $|0\rangle$ was driven by an external field activated for $\Omega t = \frac{\pi}{2}$ the Bloch vector would rotate by $\pi/2$ into an equal superposition. This is called a $\pi/2$ pulse and the plane of rotation is derived from the relative phase, ϕ , of the radiation to the system. A $\pi/2$ pulse about the x-axis is illustrated in Fig.3(c). Note high fidelity $\pi/2$ pulses with a non zero detuning can still be implemented given the detuning is small in comparison to the Rabi frequency, the same is true when the interactions of our system are small compared to the Rabi frequency.

Ramsey sequences

To validate a quantum simulator we require a measurable parameter which reflects the dynamics of the system. Ramsey interferometry utilizes two $\pi/2$ pulses separated by a time of free evolution to probe the coherences of a system. A Ramsey sequence progresses as follows. An initial $\pi/2$ pulse rotates the Bloch vector onto the equator creating a superposition of states. The system then freely evolves in time and the dipole interactions cause oscillations in the relative phases of each superposition. This translates to a rotation around the Bloch sphere equator or an oscillation in the coherences. Finally, a second $\pi/2$ pulse projects the coherences back onto the population difference of each state, represented by the z axis of the Bloch sphere, for measurement. Overall, a Ramsey pulse sequence can be summarized with its propagator

$$\begin{aligned} R_{-y}^{\frac{\pi}{2}} R_z^{\phi} R_{-y}^{\frac{\pi}{2}} &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -i \sin \phi/2 & \cos \phi/2 \\ -\cos \phi/2 & i \sin \phi/2 \end{pmatrix}, \end{aligned} \quad (27)$$

where ψ is the relative phase accumulated in the free evolution and reflects the interactions of the system. For an initial state $|0\rangle$, Eq.27 presents the population oscillation between $|0\rangle$ and $|1\rangle$ depending on the applied phase ψ . This dependence forms Ramsey fringes which are oscillations in the population difference. In the case where no phase is accumulated the two $\pi/2$ pulses add to a π pulse and the entire initial population in $|0\rangle$ transitions to $|1\rangle$ (and vice versa for the initial population in $|1\rangle$).

In our work, the second $\pi/2$ pulse is 180° out of phase with the first causing non interacting particles to end in the $|0\rangle$ state, the process is visualized in Fig.4.

Overall, the Ramsey sequence outputs experimentally accessible oscillations in the population difference if the particles interact. Therefore, these oscillations allow us to probe the evolution of the system's correlations and verify the accuracy of the same measurement done experimentally.

Large system dynamics

Generalizing the dipole dipole interaction

Eq.15 can readily be generalized to a many particle lattice where each lattice site contains zero or one particle. By rewriting each operator in their second quantized form, $\mathbb{1}_i \rightarrow \mathbf{n}_i \equiv \sum_{\sigma=0,1} \mathbf{a}_{i\sigma}^\dagger \mathbf{a}_{i\sigma}$ is now the total molecular density operator and counts the number of particles at each lattice site and

$$\mathbf{S}_i^a = \frac{1}{2} \sum_{\mu\nu} \mathbf{a}_{i\mu}^\dagger [\sigma^a]_{\mu\nu} \mathbf{a}_{i\nu}, \quad (28)$$

where $\mathbf{a}_{iv}$ destroys a particle at site i in spin state v and σ^a is a Pauli spin matrix. When there is either 0 or 1 molecule in each lattice site the term proportional to V is constant and can be dropped. Similarly, the W term generates an effective magnetic field at each lattice site which

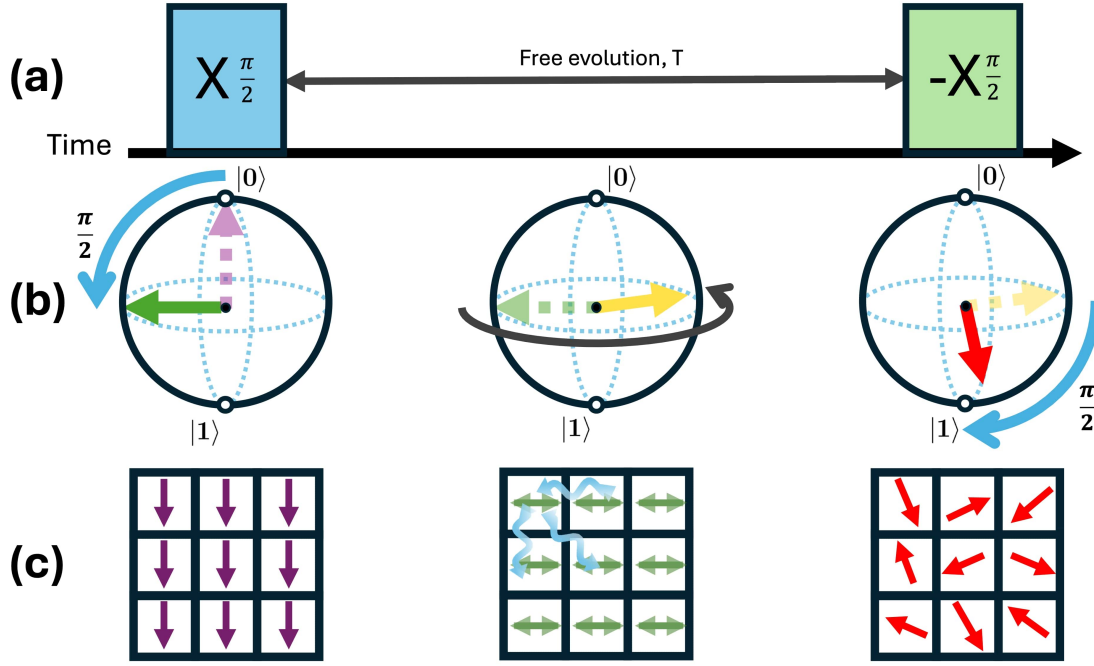


FIG. 4: Ramsey interferometry. The Ramsey pulse sequence's effect is shown in three steps. **(a)** Ramsey pulse sequence used in this work. X and $-X$ represent the axis each $\pi/2$ pulse drives the system around on the Bloch sphere. **(b)** Ramsey sequence represented on the Bloch sphere. For each step of the Ramsey sequence: dotted Bloch vectors are the initial state of the particle, solid Bloch vectors are the resultant state, a blue arrow represents the action of a $\pi/2$ pulse and a black arrow represents some free evolution of a particle. **(c)** The effect of a Ramsey sequence on a lattice of particles this work studies. All particles are initially in the ground state $|0\rangle$ (represented by $\downarrow$ to draw parallels to spin). Two arrows in one lattice site represent a superposition and a portion of the interactions are represented with blue arrows.

is uniform when edge effects are ignored. Hence, the W term commutes with the Hamiltonian and can be dropped giving

$$H = \frac{1}{2} \sum_{i \neq j} \frac{1 - 3 \cos^2 \theta_{ij}}{|\mathbf{r}_i - \mathbf{r}_j|^3} \left[\frac{J_{\perp}}{2} (\mathbf{S}_i^+ \mathbf{S}_j^- + \text{h.c.}) + J_z \mathbf{S}_i^z \mathbf{S}_j^z \right]. \quad (29)$$

Note the factor of $1/2$ eliminates double counting, thus we ignore it in later references to the couplings in Eq.29. Therefore, we have derived the XXZ spin model for a 2D lattice of pinned molecules, and we can set the electric field to zero to receive the XY Hamiltonian which we simulate. Overall, the system we simulate looks like Fig.2(a) and interacts according to Eq.29 where the Ising term vanishes, leaving the XY spin model.

Previous work

[3] measures the dipole dipole interactions of RbCs with a quantum gas microscope. This work emulates their measurements of the $|1\rangle$ population in a system of ultracold polar RbCs molecules after a Ramsey pulse, as we can classically model the results to verify the quantum

simulator is accurate. In particular, we build on their simulations which verify their experimental results by proposing an alternate method which reduces computation times and provides insight into the correlations of the system.

We use [3]’s computational method as a reference to compare to our own. They generate 2500 smaller samples of a particle distribution, simulate each under the Hamiltonian and pulse sequence studied, then average the proportion of particles in the $|1\rangle$ state for the overall systems dynamics. The result of this simulation is displayed in Fig.5, as the solid curve, where they have simulated 12 particles per sample. Also,[3] takes samples which represent a radial distribution of particles, however this work inspects uniform distributions so we adapt their method for our comparisons.

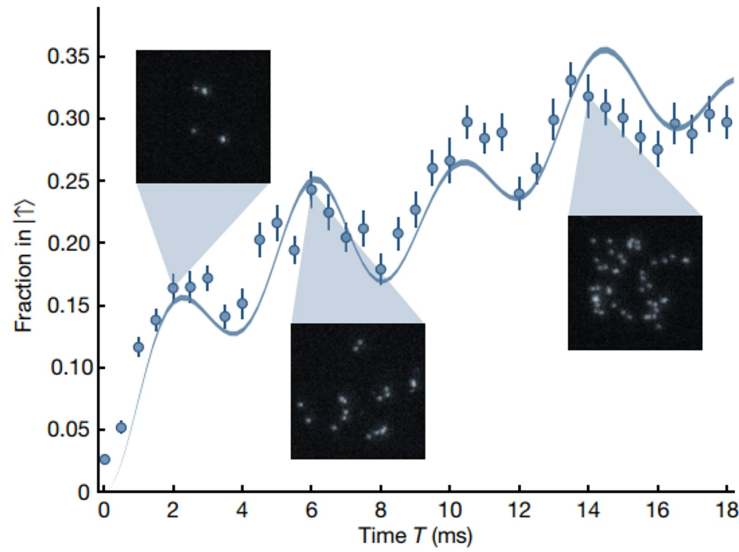


FIG. 5: Fraction of particles in up, $|1\rangle$, in a radially distributed system of molecules after a Ramsey pulse. The lattice filling is 0.054(4) and the quantization axis is through the page such that the interactions are isotropic. Insets display the site resolved $|1\rangle$ measurements at each time. Graph adapted from [3].

There are other computational methods from which we take inspiration. Disjoint cluster expansion (DCE)[6] and moving-average cluster expansion (MACE)[5] involve simulating smaller clusters of the system. However, these methods focus on specific system configurations rather than the dynamics when the distribution of particles is infinite. We look to exploit the statistical nature of the particle distribution to approximate the dynamics of a quantum system efficiently. The computational tool developed in this work therefore is suited for validating quantum simulators in classically modellable regimes.

Method

Here we present our computational method in detail which builds on previous work to simulate the dynamics of large systems of interacting particles under global pulses. Our method simulates the evolution of every possible particle configuration in an isolated 3×3 lattice under a Hamiltonian and global pulse sequences of interest, then sums the results together according to

their frequency within the relevant particle distribution. In this work, we apply this method to an infinite uniform distribution of particles with some lattice filling ρ and interactions described by the XY spin model stated in Eq.3.

Firstly, we must find every configuration of particles within a 3x3 grid. By initially finding every way to place zeros and ones in a nine-element sequence (there are well established tools for this computation) we equate each element in the sequence to a position in the 3x3 grid and take zeros and ones to represent empty and filled lattice sites respectively. Overall, this gives 512 particle configurations of a 3x3 lattice. However, if we are studying a Hamiltonian featuring purely isotropic interactions then configurations connected by rotations will have the same dynamics and we can group these into orbits (as the groups are self contained under the symmetry operations we consider). Consequently, only one representative from each orbit needs to be simulated significantly reducing the computational load.

To sort each configuration into their respective orbits we first apply every symmetry operator, under which the dynamics are invariant, to each configuration. For a 3x3 grid this translates to four 90 degree rotations about the center, a reflection through the center parallel to one side and translations within the grid. Additionally, combining the symmetry operators and applying each to every configuration generates the full orbits for all 512 configurations. However, since the initial configurations have equal orbits to others, duplicate orbits will be generated. Removing these duplicates will leave the configurations sorted into their correct orbits.

We remove duplicate orbits as follows. By subtracting each generated group member from the initial configuration (acted upon by the symmetry operators) then checking the value of the first non zero entry in the result we can either keep or discard the group. If the value found is -1 then the group has already been generated and we immediately discard it. If we find 1 then the group may not have been generated and we continue to check the other group members. If there is no non zero values then the group is symmetrical under the specific transformation checked and we continue to check the other group members. Finally, if every group member does not have -1 as the first non zero entry when subtracted from the initial configuration then the group has not been generated and we keep it (with the initial configuration as the representative). This process can also be viewed similarly to lexicographical ordering. Assigning a letter to each lattice site alphabetically (top left is a, top middle is b e.t.c), each configuration will produce a sequence of letters. Only the configuration we choose as the representative will alphabetically come first in its group, therefore any group generated from a member which is not first alphabetically can be discarded as a duplicate.

Additional simplifications can be made depending on what is observed and the pulse sequence implemented. The simulations in this work count the population of particles in the $|1\rangle$ state, however the Ramsey pulse (with a pulse phase difference, Fig.4) used in this work applied to a single isolated particle results in a pure $|0\rangle$ state. Therefore, for this work, we can ignore the dynamics of configurations only featuring a single particle.

Overall, the 84 representatives found for each group of 3x3 particle configurations with relevant dynamics when the interactions are isotropic are presented in Fig.6. We simulate each ones Ramsey fringes to generate the overall Ramsey fringes of a large system.

By finding the proportion of particles in each configuration for the large particle distribution studied we can construct the system from its constituent parts. Also, the constituent 3x3 particle configurations must be isolated for the dynamics to be localized within them. We interpret this

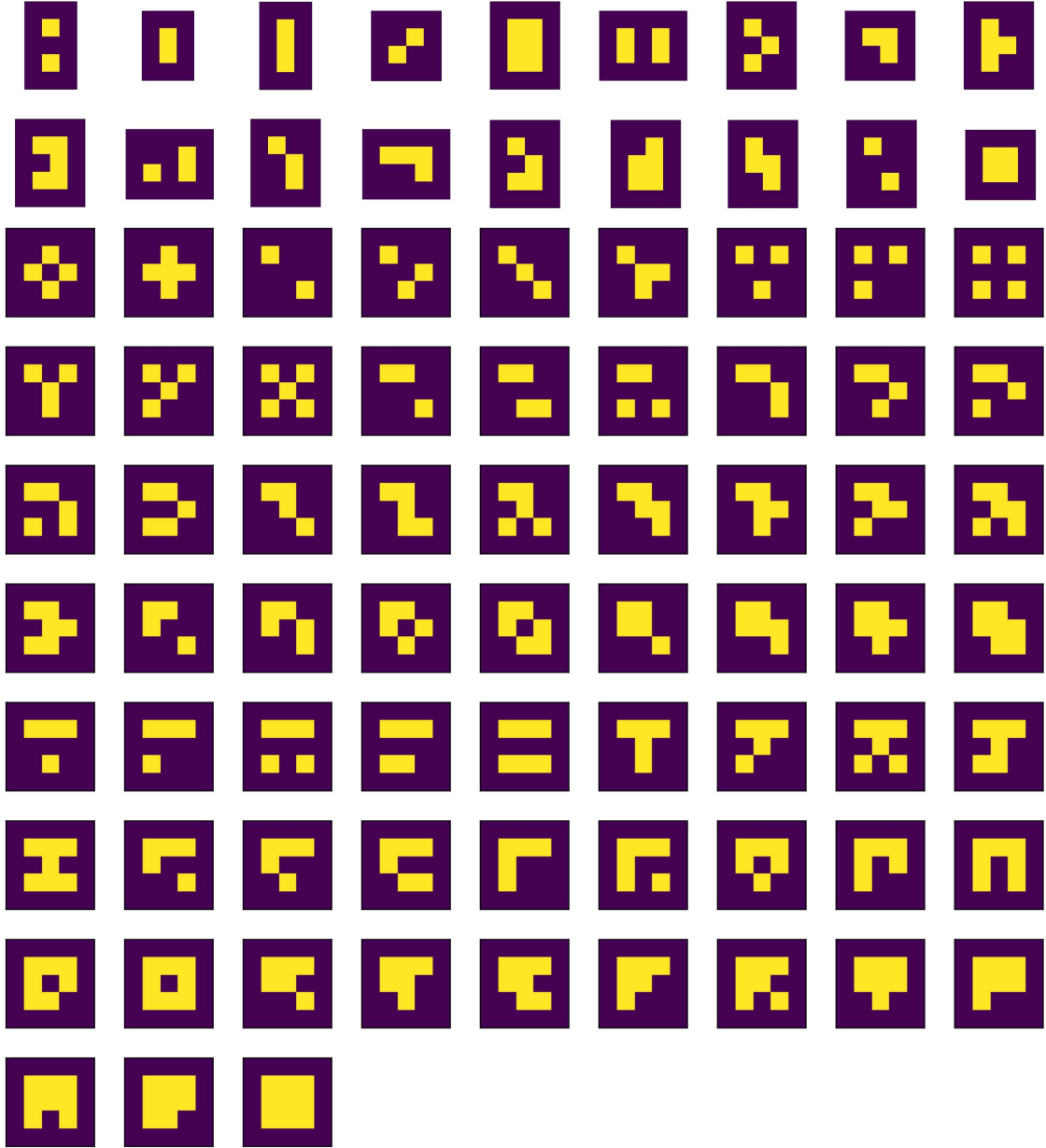


FIG. 6: Particle configurations of a 3x3 grid with isomorph rejection. Yellow squares represent particles and dark squares represent empty lattice sites. Configurations connected by rotations or reflections not included due to identical isotropic interactions.

to mean that each configuration has a 'border' of empty sites around it.

This work simulates a large uniform distribution with some lattice filling, ρ . Therefore, by considering the chance a particle configuration appears at each lattice point and integrating over the lattice where the probability a particle occupies a lattice site is ρ , we can write the proportion of particles within some configuration (or other member of its orbit) analytically as

$$f = S(1 - \rho)^e \rho^{o-1}, \quad (30)$$

where e and o are the number of empty and occupied sites in the configuration and S is the number of configurations in its orbit. Eq.30 is plotted for each representative configuration

(Fig.6) in Fig.7(b).

Fig.7(c) displays the particles which are unaccounted for in our method at each lattice filling. This justifies why we consider isolated 3×3 grids since at the lattice fillings we consider (about 5% to 15%) we can encapsulate the majority of the system with configurations of this size and implementing larger configuration sizes would increase the computational load unnecessarily.

For different particle distribution (e.g. a radial distribution) a different method is required to find the proportion of particles in each configuration. This work only presents results for a uniform distribution however, for completeness, we briefly outline a numerical method to handle other situations. Generating random samples of some arbitrary distribution, then applying a discrete convolution with a configuration as the kernel can determine the number of times the configuration appears in the sample which, divided by the total number of particles in the sample, gives the proportion of particles in the configuration. Averaging the result over every sample system produces the proportion of particles in some configuration within the larger particle distribution studied. This numerical method can also be utilized to decompose a single system with specified particle positions rather than the infinite distribution this work inspects.

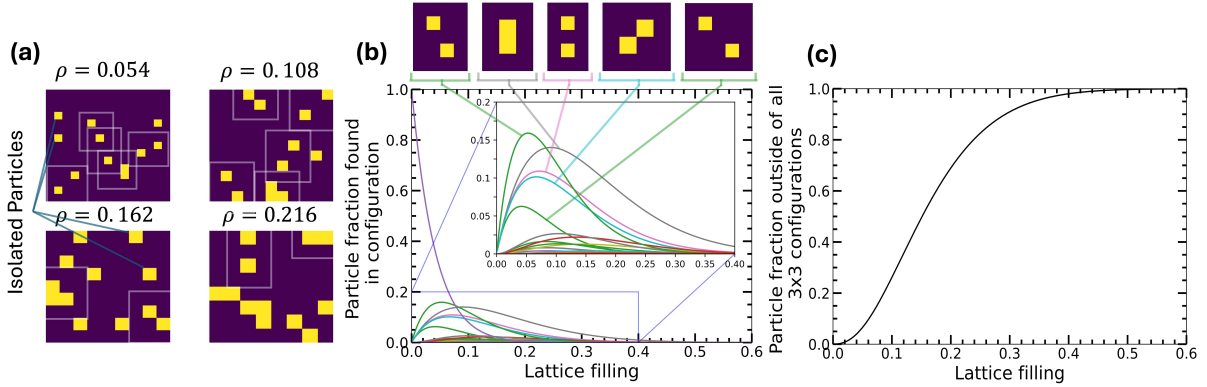


FIG. 7: Particle fraction found in each 3×3 isolated configuration for a uniform distribution at each lattice filling. (a) Example uniform particle distributions with their component 3×3 configurations labeled. Isolated particles do not contribute to the dynamics. (b) Proportion of particles found in each orbit, Fig.6, in an infinite uniform distribution at each lattice filling. The purple line represents isolated particles. (c) Proportion of particles which are not in an isolated 3×3 configuration at each lattice filling.

To obtain our total values for the system, we must simulate the evolution of each individual configuration under the Hamiltonian and pulse sequence of interest. Then we combine the Ramsey fringes according to the proportion of particles within each configuration to give the total dynamics. This work simulates the XY spin model, Eq.3, under a Ramsey pulse (with a phase difference of $\pi/2$ between pulses) for comparison with [3]. Therefore, we simulate every system depicted in Fig.6 with interaction strengths $J_{\perp}/2 = 1500 \text{ Hz}$, 750 Hz and lattice fillings $\rho = 0.054, 0.108, 0.162$ chosen to be close to current RbCs experiments. Additionally, we explore the change when the quantization axis is in the plane of the particles parallel to an axis creating an anisotropic interaction (top row Fig.1(b) and (c)).

A subset of the Ramsey fringes calculated are shown in Fig.8. Interestingly, configurations with an odd number of particles have their Ramsey fringes 'capped' implying that there are significant topological effects at play. However, this is beyond the scope of this work so is

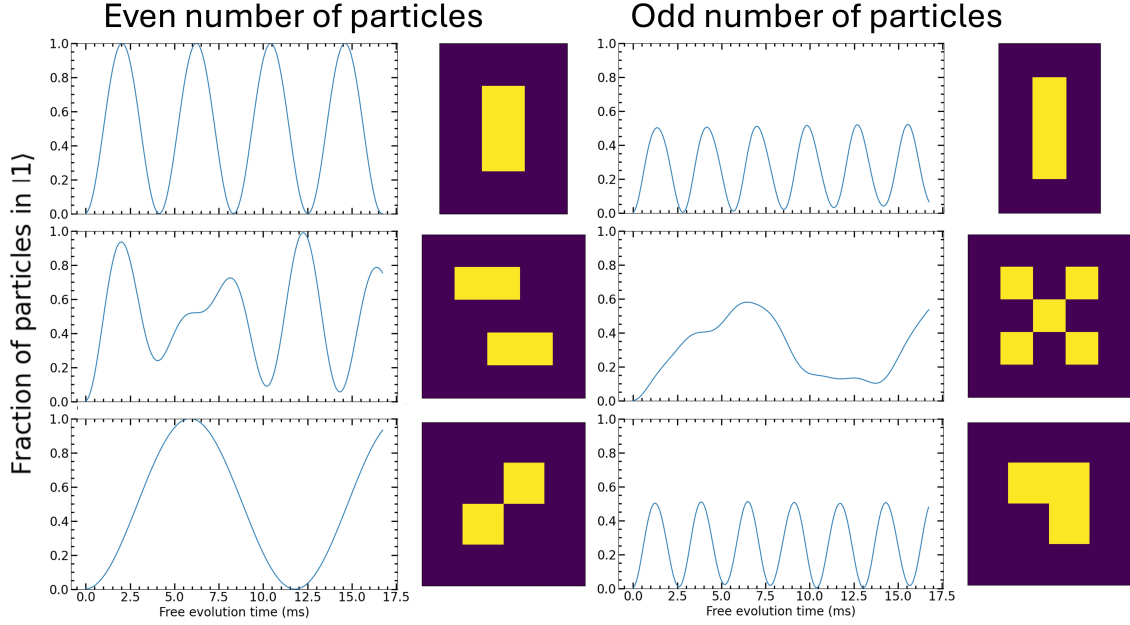


FIG. 8: Example of unweighted Ramsey fringes of 3x3 configurations. The interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz and the quantization axis points into the page to create an isotropic interaction. Configurations containing an even and odd number of particles are distinguished by row to reflect their differing features.

designated for future study.

Multiplying each component Ramsey fringe with the proportion of particles in the corresponding configuration and then summing for all configurations gives the total dynamics of the system. The final weighted component Ramsey fringes are illustrated in Fig.9.

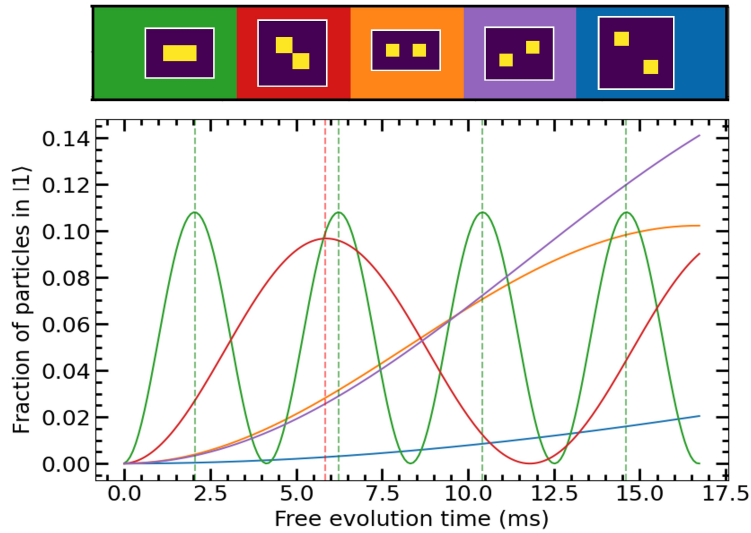


FIG. 9: Weighted component isotropic Ramsey fringes. Component Ramsey fringes of Fig.6 for an interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz and lattice filling 0.054. Each plot is color coded with its corresponding configuration.

Another simplification which is not utilized in this work but can be made to improve our

programs efficiency, is to ignore configurations which make up a very small proportion of the overall system. For example, in a uniform distribution with any lattice filling, Eq.30 indicates the last configuration displayed in Fig.6 makes up a negligible part of the system as it is extremely unlikely to find nine particles in a square surrounded by empty sites. Therefore, this configuration can be removed from the calculations with no significant effect on the results. Also, since the other configurations which appear sparsely within a uniform distribution contain a larger amount of particles, the time required to simulate their individual dynamics is exponentially larger than for the relevant configurations. Consequently, the computational load can be cut significantly with this approximation, however this depends on the particle distribution studied.

Additionally, Fig.7(a) illustrates the double counting of particles where a particle is classified into several configurations. This effect is reflected within Eq.30 and can be minimized by subtracting the particle fraction of a complex configuration from the configuration contained inside it (while scaling to account for the number of particles in each configuration). However, we found double counting has a minimal effect on our results likely because the dominant components of the dynamics are nearest and next-nearest neighbor pairs and the majority of the particles counted twice are surrounded by empty sites. Consequently, their interactions are too weak to significantly impact our results. Also, the number of particles unaccounted for, Fig.7(c), is never negative as would be the case if double counting was significant. Although, this effect may become significant when simulating a particle distribution built from configurations which are spaced further apart.

Results

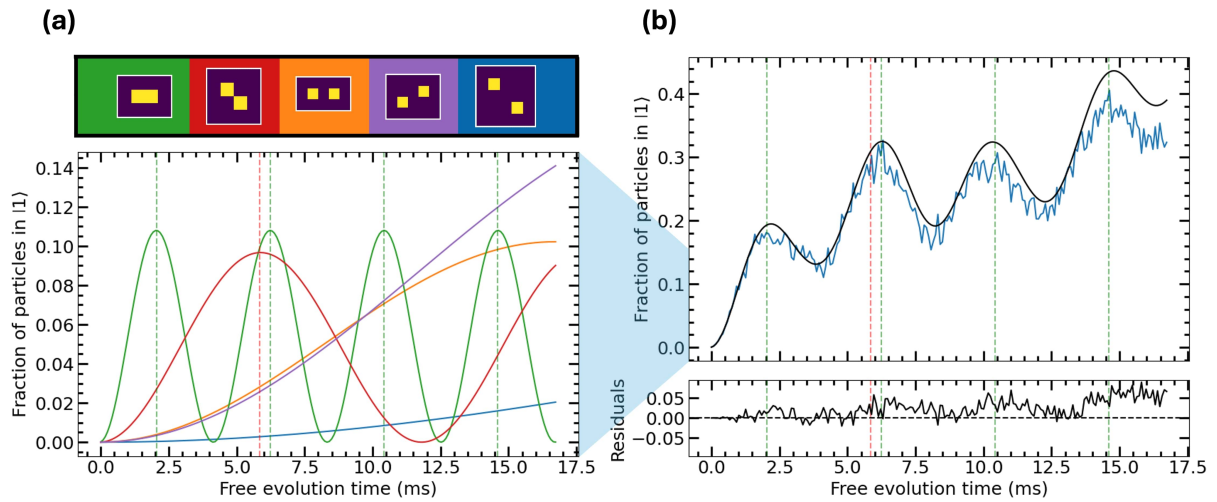


FIG. 10: Ramsey interferometry of an infinite uniform system. The interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz and the lattice filling is 0.054. **(a)** The largest contributing component Ramsey fringes are shown and each configuration is color coded with its corresponding plot. **(b)** Overall Ramsey fringe for an infinite uniform particle distribution. The same system simulated with [3]’s method is included for comparison (blue line), each point was sampled 50 times with 12 particles.

Here we present our results for a system with an isotropic interaction and interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz under a Ramsey pulse sequence. From Fig.10 we can extract the average dynamics of a quantum system efficiently without noise.

The population oscillations displayed in Fig.10 have been observed experimentally in previous work[3] validating our results. Our simulation has significant agreement with previous calculations (blue line) suggesting that our assumption that the dynamics are localized in 3×3 lattices is valid at this lattice filling, interaction strength and timescale. Consequently, we can infer that our simulation does not break down from edge effects under these conditions. With experimentally accessible dynamics and our efficient and accurate computational method at 0.054 lattice filling, $\frac{J_{\perp}}{2} = 1500$ Hz interaction strength and 20 ms timescale we conclude that this measurement is ideal for verifying the accuracy of a quantum simulator under the XY spin model.

The reference simulation (blue line) features significant noise compared to Fig.5. This is due to sampling each point 50 times, to cut computation times, rather than 2500 times as shown in Fig.5. So the result is more sensitive to sampling.

The individual component weighted Ramsey curves are displayed in Fig.10(a). We observe the highest contribution to the dynamics in this timescale is from nearest neighbor pairs. This is expected as the overall Ramsey fringe is dependent on the correlations of the particles. Therefore, since nearest neighbor particles will have the strongest interactions and they are common at low lattice fillings (Fig.7(b)) they will have a dominant contribution in the small timescale we consider. Additionally, we observe that diagonal pairs, the second strongest interaction among the configurations, have the second largest contribution for the same reasons. The nearest neighbor pairs create the three peaks in the results and the diagonal pairs elevates the middle peak higher, therefore the strength of the interaction in these configurations is clearly displayed and could be inferred for similar oscillations.

Fig.10(a) also conveys that pair configurations dominate the dynamics. At low lattice fillings it is expected that configurations with less particles will comprise more of the overall system. An isolated pair of particles separated by r have a state $\sin(\frac{J}{4\pi r^3}T) |00\rangle - i \cos(\frac{J}{4\pi r^3}T) |11\rangle$ after a Ramsey pulse with a free evolution time of T . Therefore, at low lattice fillings, the proportion of particles in $|1\rangle$ is well approximated by sum of out of phase (determined by interaction strength) sine functions created by each of the nearest neighbor pairs of ascending order.

Apart from the highest two contributors to our calculations in Fig.10, we note the increase in the proportion of $|1\rangle$ states from the component Ramsey fringes with time. This reflects the increasing correlations as the system evolves. Ramsey fringes of configurations with weaker interactions have a larger period as their interactions drive transitions slower (see pair state equation in previous paragraph). Consequently, their contributions cause a gradual increase the the proportion of $|1\rangle$ states in the system.

While our simulation is in good agreement with the reference simulation, the residuals in Fig.10(b) reveal a slight overestimation in the population of $|1\rangle$ states. This exists because our method neglects edge effects more than the reference method. Edge effects arise from particles in the periphery of our configurations which we ignore. An advantage of simulating more particles at a time is that more periphery particles are accounted for and therefore edge effects will be mitigated.

Fig.11 illustrates why our simulation overestimates the fraction of $|1\rangle$ states with time. We

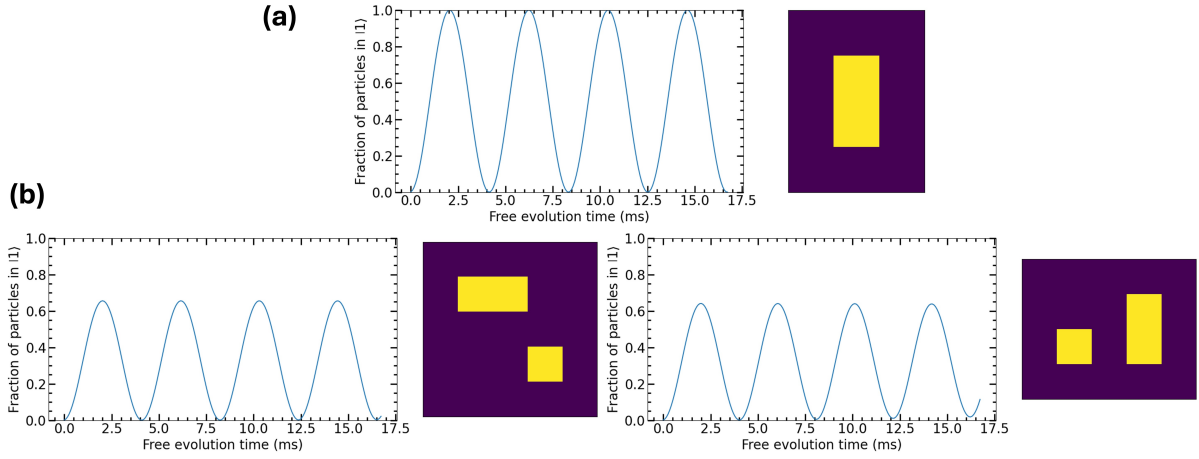


FIG. 11: Unweighted Ramsey fringes of a nearest neighbor pair with and without periphery particles. The interaction strength $J/2 = 1500$ Hz and the interactions are isotropic. **(a)** Unweighted Ramsey fringe of nearest neighbor pairs. **(b)** Unweighted Ramsey fringes of a nearest neighbor pair with a periphery particle.

know from Eq.30 and Fig.7 that a uniform distribution is composed primarily from close pairs at low lattice fillings. Therefore, while we approximate each configuration as isolated, it is common for periphery particles, outside of the 'border' of empty sites, to have an effect on the evolution of the pairs. Fig.11 demonstrates that introducing another particle to a pair configuration 'caps' the maximum proportion of $|1\rangle$ states detected. At very low timescales (less than half a period of its dynamics) this has a minimal effect however as the system evolves our calculations will not 'cap' the $|1\rangle$ population of the configuration and we will overestimate. This effect could be mitigated by considering particle configurations beyond 3x3 to partially account for particles on the edge.

Higher lattice fillings and different interaction strengths

Fig.12 indicates how increasing the lattice filling effects our simulations. We infer that our simulations break down beyond lattice fillings of 0.108 as our simulations diverge from the reference model as we increase the filling, shown by the residuals in Fig.12. Also, as we increase the lattice filling past 0.108 our Ramsey fringes decrease overall which is counter to what we would expect as the system would be more correlated. This breakdown can be attributed to the particles which are unaccounted for, shown in Fig.7(c), which becomes significant at these lattice fillings. Therefore, the accuracy of these simulations could be improved by considering configurations beyond 3x3 thus accounting for a larger proportion of the overall system.

Now consider the Ramsey fringes when the interaction strength is halved, shown in Fig.13, to $\frac{J_{\perp}}{2} = 750$ Hz. This could be achieved in ultracold molecules by choosing a different set of states to encode spin 1/2 or by level dressing the rotational eigenstates with microwave radiation.

As expected the period of the dynamics have doubled, resulting in lower levels of entanglement. Also, the Ramsey fringe retains the same features as Fig.10. Consequently, our simulation significantly agrees with the reference simulation and we conclude that this measurement could

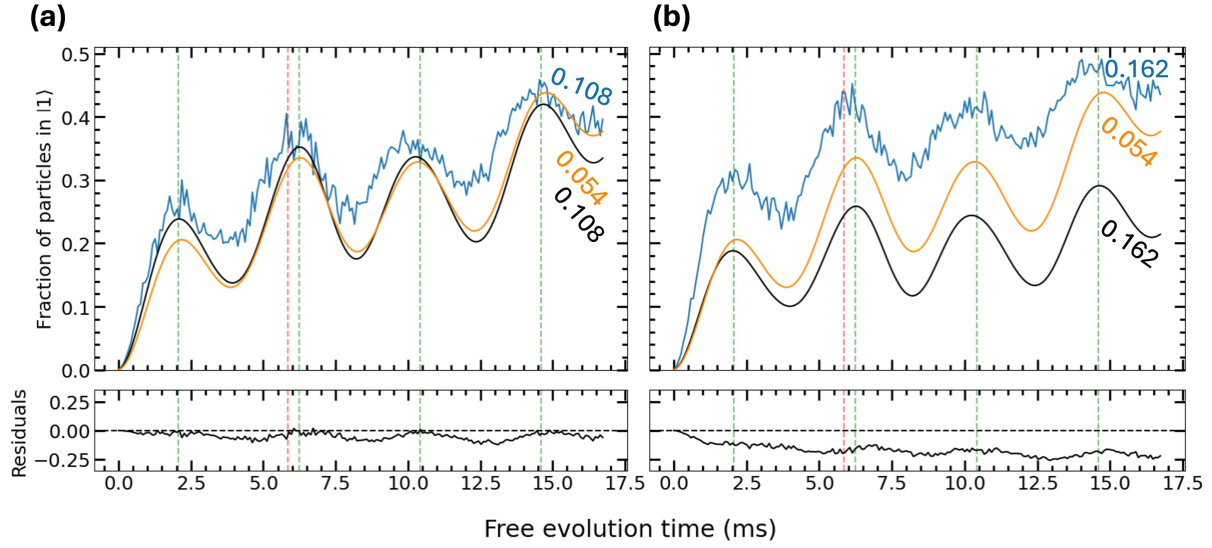


FIG. 12: Ramsey interferometry of higher uniform lattice fillings. The interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz. The same systems simulated with [3]’s method are included for comparison for both lattice fillings (blue lines), each point was sampled 50 times with 12 particles. The result for a lattice filling of 0.054 is included on both plots for comparison (orange line). **(a)** Overall Ramsey fringe for a uniform infinite particle distribution with a lattice filling of 0.108. **(b)** Overall Ramsey fringe for a uniform infinite particle distribution with a lattice filling of 0.162.

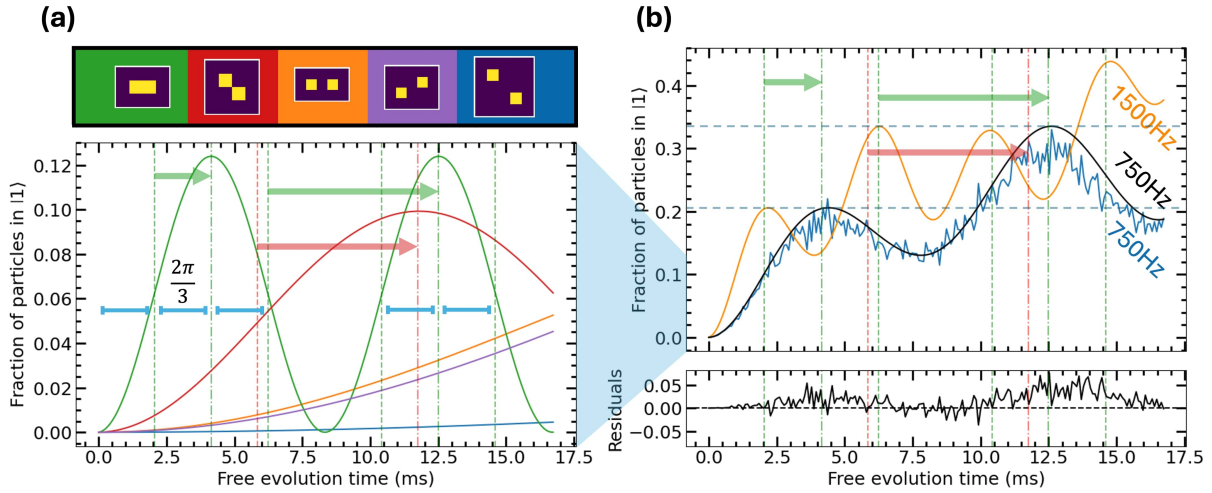


FIG. 13: Ramsey interferometry of an infinite uniform system with a halved interaction strength.

Lattice filling is 0.054. Peaks of component Ramsey fringes at $\frac{J_{\perp}}{2} = 1500$ Hz are included for comparison **(a)** The largest contributing component Ramsey fringes. The key is color coded for each configuration and corresponding plot. **(b)** Overall Ramsey fringe of a uniform infinite system for an interaction strength of $\frac{J_{\perp}}{2} = 750$ Hz. The same system simulated with [3]’s method is included for comparison (blue line), each point was sampled 50 times with 12 particles. The orange line shows the Ramsey fringe when $\frac{J_{\perp}}{2} = 1500$ Hz.

be utilized for verification of quantum simulators. We expected this as the correlations are low therefore the residuals in Fig.13(b) are minimal.

Anisotropic

Now we set the quantization axis parallel with the plane of particles rather than through, which can be implemented with a magnetic field. From Eq.6 and Fig.1 we know this gives rise to an anisotropic interaction between particles. The anisotropic interaction strength around a particle is presented in Fig.1.

Modifying our computational method we simulate the component Ramsey curves with an anisotropic interaction. This breaks the symmetry of our 3x3 grid, therefore the configurations cannot be isomorphically rejected and sorted into orbits. As a result, we simulate the evolution of all 512 particle configurations before performing the weighted sum for the overall dynamics. The results are displayed in Fig.14.

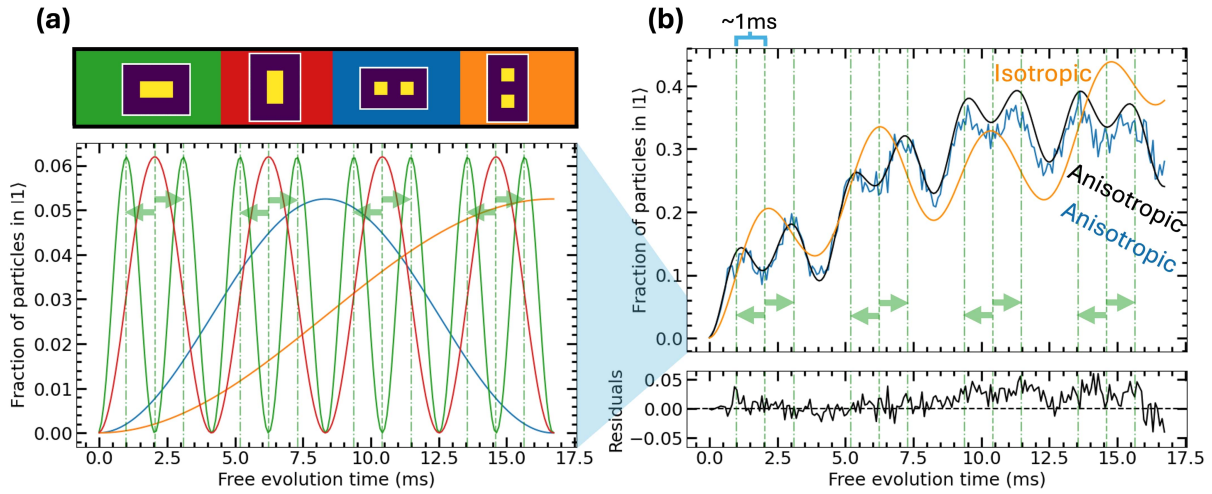


FIG. 14: Ramsey interferometry of an infinite uniform system with anisotropic interactions. The quantization axis points in the plane of the particles parallel to one of the edges. The interaction strength $\frac{J_{\perp}}{2} = 1500$ Hz and the lattice filling is 0.054. The peaks of the component Ramsey fringes for isotropic interactions are displayed for comparison. **(a)** The largest component Ramsey fringe contribution curves are displayed and color coded with their respective configurations. **(b)** Ramsey fringes of an infinite uniform system with anisotropic interactions. The same system simulated with [3]’s method is included for comparison (blue line), each point was sampled 50 times with 12 particles. The Ramsey fringe for isotropic interactions are shown for comparison (orange line).

We distinguish a ‘splitting’ of the peaks compared to the isotropic case, Fig.10. From the component Ramsey curves displayed, we attribute this to the ‘splitting’ of the component curves as the dynamics of a rotated configuration features enhanced or suppressed interaction strengths between particles. In the case of nearest neighbor pairs, this means horizontal pairs have double the interaction strength compared to vertical pairs (and the isotropic case). Consequently, their Ramsey fringes period is halved creating the three double peaks. For diagonal pairs, the quantization axis makes the same angle with both rotations of the configuration. Consequently, the interaction strength between the particles is halved and the contribution of diagonal pairs has a negligible significance removing the enhancement of the second peak as in Fig.10.

Our simulation is in significant agreement with the reference method. Fig.14 reports the highest accuracy in this work as its residuals, Fig.14(b) are the smallest. Likely, this is because

the interactions are suppressed along the $\theta = 54.7^\circ$ lines in those four directions while the interaction is enhanced in two directions. Hence, the overall entanglement of the system is lower, the dynamics are localized and so our simulation is accurate. Overall, this supports anisotropic measurements for verification of quantum simulators and implies that simulations with larger interaction strengths and longer timescales will be more robust than the isotropic case.

Conclusion and outlook

We presented a novel computational method for simulating infinite two dimensional interacting systems of particles. Building off previous work which simulated localized clusters we focused on calculating experimentally accessible parameters for an infinite system efficiently. We calculated the spin interactions according to the spin 1/2 XY model. Our work was considered in the context of an analogue quantum simulator built with ultracold polar RbCs molecules and we reviewed how to create a pseudo spin 1/2 system with this platform by encoding degrees of freedom in the rotational levels of the molecule. Also, driving transitions in the particles provided access to quantum correlations with Ramsey pulse sequences. Assuming the dynamics of the system were localized we decomposed the system into its constituent parts and modeled these parts separately to avoid exponentiating large Hilbert spaces originating from large systems of entangled particles. We extended our computations to different interaction strengths, particle densities and angular dependence of the interactions to test the limits of our work.

Our model was compared against a previous simulation, which samples a distribution of particles, for verification of our results. In regimes with low entanglement, our simulation proved sufficiently accurate and efficient for studying spin models compared to previous simulations which are computationally burdensome. However, other computational methods are more accurate as the system becomes more entangled, as we only consider 3x3 clusters of particles, due to the prevalence of edge effects and a better representation of the overall distribution. By modeling an anisotropic interaction we saw that the oscillations in the Ramsey fringes were split into two due to the breaking of rotational symmetry. Also, the suppression of interactions in the anisotropic case enabled us to obtain an especially accurate Ramsey fringe. We found our method was valid up to a lattice filling of about 0.108 and functioned accurately in regimes which ultracold polar molecules lie ($\frac{J_\perp}{2} \sim 1000$ Hz). Hence, our simulations are a useful tool for verifying the accuracy of quantum simulators and understanding electron spin interactions in a crystal for isotropic and anisotropic interactions.

All code and animations of our data is available on GitHub: <https://github.com/J-Coward/DipolarSpin>.

Apart from the directions mentioned previously in this work, there are numerous future directions for this work. A wider range of Hamiltonian could be simulated. For example, it is straightforward to simulate the XXZ Hamiltonian instead of the XY Hamiltonian as we simply add an Ising term to Eq.3. Also, by switching the system back and forth between two Hamiltonians, Floquet engineering could be explored and therefore verified in experimental systems. [3] already implements Floquet engineering in quantum simulator consisting of ultracold polar RbCs. Additionally, the nuclear spins of a molecule could encode extra degrees of freedom and

the t-J model could be investigated which allows particles to tunnel between lattice sites[2].

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Summary for a general audience

Since the discovery of quantum mechanics in the early 20th century, it has become vital to our understanding of the world and way of life. Quantum mechanics is the study of very small particles and how they interact. Magnetism, superconductivity and nuclear fission result from quantum interactions and are essential for many technological advancements such as MRI machines and nuclear generators. Furthermore, computers that operate using quantum mechanics offer an unprecedented level of computational power in the near future, inspiring impactful research into the quantum world.

Using a classical computer, this work explores how a particle's internal energy state, called spin, interacts with the internal energy state of other particles, causing magnetism. We simulate how these particles trade energy, through radiation, to swap spin states, and the impact of this happening on a large scale in a two dimensional plane. Through these trades in spin state, the state of a particle becomes dependent on the state of another particle, therefore entangling them. Exponentially more information is required to describe entangled states than unentangled states. When applied to the next generation of quantum computers, more information can be stored by entangled particles than by classic computer bytes. In practice, our work can be compared to the results of quantum simulators, based on the same spin swapping interactions, to verify they are operating correctly.

Appendix A: Tensor algebra results

In this appendix, we present formulas used in the work to resolve equations involving tensors included in the molecular part of the work.

Following [2], let $T_{p_i}^{k_i}(\mathbf{A}_i)$ be a tensor of rank k_i with components p_i which operates on angular momentum $\mathbf{J}_i$ with $i = 1, 2$. We define the tensor product of $T_{p_1}^{k_1}(\mathbf{A}_1)$ with $T_{p_2}^{k_2}(\mathbf{A}_2)$ as

$$T_p^k(\mathbf{A}_1, \mathbf{A}_2) = \sum_{p_1} T_{p_1}^{k_1}(\mathbf{A}_1) T_{p-p_1}^{k_2}(\mathbf{A}_2) (2k+1)^{\frac{1}{2}} \begin{pmatrix} k_1 & k_2 & k \\ p_1 & p-p_1 & -p \end{pmatrix} (-1)^{-k_1+k_2-p}, \quad (31)$$

where the 2 x 3 matrix is the 3j symbol. For $k_1 = k_2 = k$ we can write the scalar product of $T^{k_1}(\mathbf{A}_1)$ and $T^{k_2}(\mathbf{A}_2)$ as

$$T^{k_1}(\mathbf{A}_1) \cdot T^{k_2}(\mathbf{A}_2) = \sum_p (-1)^p T_p^k(\mathbf{A}_1) T_{-p}^k(\mathbf{A}_2). \quad (32)$$

Note when $k = 1$ the spherical and Cartesian scalar products are equal, $T^1(\mathbf{A}_1) \cdot T^1(\mathbf{A}_2) = \mathbf{A}_1 \cdot \mathbf{A}_2$.